

CRTLPyme - Deployment Guide

Overview

This guide provides multiple methods to deploy CRTLPyme to Google Cloud Run.

Target URL: <https://crtlpyme-app-399088129827.us-central1.run.app/>

GCP Project: crtlpyme-app

Region: us-central1

Account: crtlpyme@gmail.com

Prerequisites

Before deploying, ensure you have:

1.  **Google Cloud Project** - Project ID: `crtlpyme-app`
 2.  **Cloud Run API** enabled
 3.  **Cloud Build API** enabled
 4.  **Artifact Registry API** enabled
 5.  **Secret Manager** configured with required secrets
 6.  **Database** - PostgreSQL running at 136.116.45.158:5432
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Method 1: Deploy via GCP Console (Recommended for Quick Deploy)

This is the easiest method that doesn't require local gcloud installation.

Steps:

1. **Open GCP Console**
 - Go to <https://console.cloud.google.com/>
 - Login with crtlpyme@gmail.com
 - Select project: `crtlpyme-app`
2. **Navigate to Cloud Build**
 - Go to: <https://console.cloud.google.com/cloud-build/builds>
 - Or search for "Cloud Build" in the top search bar
3. **Trigger a Build**
 - Click "RUN BUILD" button
 - In "Source" section, select "Repository" → "Connect Repository"
 - Connect your GitHub repository: `CRTLPyme`
 - Select branch: `main`
 - Build configuration: `Cloud Build configuration file (cloudbuild.yaml)`
 - Click "RUN BUILD"

4. Monitor Deployment

- Watch the build logs in real-time
- Build takes approximately 5-10 minutes
- Wait for "SUCCESS" status

5. Verify Deployment

- Navigate to Cloud Run: <https://console.cloud.google.com/run>
- Find service: `crtlpyme`
- Click on the service URL to test
- Expected URL: <https://crtlpyme-app-399088129827.us-central1.run.app/>

Method 2: Deploy via GitHub Actions (Recommended for CI/CD)

Automated deployment on every push to the main branch.

Setup Steps:

1. Create Service Account

```
```bash
In GCP Console or Cloud Shell, run:

Create service account
gcloud iam service-accounts create github-actions \
 --display-name="GitHub Actions Deployer"

Grant necessary permissions
gcloud projects add-iam-policy-binding crtlpyme-app \
 --member="serviceAccount:github-actions@crtlpyme-app.iam.gserviceaccount.com" \
 --role="roles/run.admin"

gcloud projects add-iam-policy-binding crtlpyme-app \
 --member="serviceAccount:github-actions@crtlpyme-app.iam.gserviceaccount.com" \
 --role="roles/cloudbuild.builds.builder"

gcloud projects add-iam-policy-binding crtlpyme-app \
 --member="serviceAccount:github-actions@crtlpyme-app.iam.gserviceaccount.com" \
 --role="roles/iam.serviceAccountUser"

Create and download key
gcloud iam service-accounts keys create key.json \
 --iam-account=github-actions@crtlpyme-app.iam.gserviceaccount.com
```
```

1. Add Secret to GitHub

- Go to your GitHub repository
- Settings → Secrets and variables → Actions
- Click "New repository secret"
- Name: `GCP_SA_KEY`
- Value: Paste the entire contents of `key.json`
- Click "Add secret"

2. Commit and Push Workflow

```
bash
cd /home/ubuntu/github_repos/CRTLPyme
git add .github/workflows/deploy-to-cloud-run.yml
git commit -m "Add GitHub Actions deployment workflow"
git push origin main
```

3. Trigger Deployment

- Method A: Push to main branch (automatic)
- Method B: Manual trigger via GitHub UI
 - Go to Actions tab
 - Select "Deploy to Cloud Run"
 - Click "Run workflow"

Method 3: Deploy via Local gcloud CLI

For developers with gcloud installed locally.

Prerequisites:

```
# Install gcloud CLI
curl https://sdk.cloud.google.com | bash
exec -l $SHELL

# Initialize gcloud
gcloud init
```

Deployment Steps:

1. Authenticate

```
bash
gcloud auth login
gcloud config set project crtlpyme-app
```

2. Submit Build

```
bash
cd /home/ubuntu/github_repos/CRTLPyme
gcloud builds submit --region=us-central1
```

3. Monitor

- Follow build logs in terminal
- Or visit: <https://console.cloud.google.com/cloud-build/builds>

Alternatively, use the provided script:

```
chmod +x /home/ubuntu/crtlpyme_deploy_cloudbuild.sh
/home/ubuntu/crtlpyme_deploy_cloudbuild.sh
```

Required GCP Secrets

Ensure these secrets are configured in Secret Manager:

| Secret Name | Description | Example Value |
|-------------------------|------------------------------|---|
| DATABASE_URL | PostgreSQL connection string | postgresql://
user:pass@host:5432/db |
| NEXTAUTH_SECRET | NextAuth.js secret key | Random 32+ char string |
| SENDGRID_API_KEY | SendGrid API key | SG. xxxxxxxxxxxx |
| SENDGRID_FROM_EMAIL | Sender email address | noreply@crtlpyme.cl |
| transbank-api-key | Transbank API key | Your Transbank key |
| transbank-commerce-code | Transbank commerce code | Your commerce code |
| TRANSBANK_ENVIRONMENT | Transbank environment | production or integration |

How to Add Secrets:

1. Go to Secret Manager: <https://console.cloud.google.com/security/secret-manager>
2. Click "CREATE SECRET"
3. Enter secret name and value
4. Click "CREATE"

Repeat for all required secrets.

Environment Variables

The following environment variables are automatically set by Cloud Run:

```

NODE_ENV: production
NEXTAUTH_URL: https://crtlpyme-vhndaaajwpq-uc.a.run.app
NEXT_PUBLIC_APP_NAME: CRTLPyme
GOOGLE_CLOUD_PROJECT_ID: crtlpyme-app

```

Database Configuration

Connection Details:

- Host: 136.116.45.158
- Port: 5432
- Database: crtlpyme
- User: postgres
- Password: CRTLPyme2025!

DATABASE_URL Format:

```
postgresql://postgres:CRTLPyme2025!@136.116.45.158:5432/crtlpyme
```

⚠ Important: Store this in Secret Manager, never in code or environment variables directly.

Troubleshooting

Build Fails

1. **Check Build Logs**
 - Go to Cloud Build console
 - Click on the failed build
 - Review error messages
2. **Common Issues:**
 - Missing secrets → Add to Secret Manager
 - Insufficient permissions → Check service account roles
 - Database connection → Verify database is accessible
 - Memory issues → Increase Cloud Run memory allocation

Application Won't Start

1. **Check Cloud Run Logs**

```
bash
gcloud logging read "resource.type=cloud_run_revision AND resource.labels.service_name=crtlpyme" --limit 50 --format json
```
2. **Or via Console:**
 - Go to Cloud Run → crtlpyme service
 - Click "LOGS" tab
 - Look for startup errors

Database Connection Issues

1. **Verify Database Access**

```
bash
psql -h 136.116.45.158 -U postgres -d crtlpyme
```
2. **Check Cloud Run Egress**
 - Ensure Cloud Run can access external IPs
 - May need VPC Connector for private IPs

Secrets Not Loading

1. **Verify Secret Exists**
 - Go to Secret Manager
 - Check secret names match exactly (case-sensitive)
 2. **Check Service Account Permissions**
 - Service account needs `roles/secretmanager.secretAccessor`
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Post-Deployment Checklist

- [] Application loads at <https://crtlpyme-app-399088129827.us-central1.run.app/>
 - [] Database connection is successful
 - [] User login works
 - [] Transbank payment integration works (test transaction)
 - [] Email notifications work (SendGrid)
 - [] SSL certificate is valid
 - [] Health check endpoint responds
 - [] API routes are accessible
 - [] Static assets load correctly
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Monitoring & Maintenance

View Logs

```
gcloud logging read "resource.type=cloud_run_revision" --limit 50
```

View Metrics

- CPU usage: <https://console.cloud.google.com/run/detail/us-central1/crtlpyme/metrics>
- Memory: Same URL, select “Memory” tab
- Request count: Same URL, select “Requests” tab

Update Service

```
# Update environment variable
gcloud run services update crtlpyme \
  --region=us-central1 \
  --set-env-vars=NEW_VAR=value

# Update secret
gcloud run services update crtlpyme \
  --region=us-central1 \
  --update-secrets=SECRET_NAME=secret-name:latest
```

Rollback

If deployment fails or causes issues:

```
# List revisions
gcloud run revisions list --service=crtlpyme --region=us-central1

# Rollback to previous revision
gcloud run services update-traffic crtlpyme \
  --region=us-central1 \
  --to-revisions=REVISION_NAME=100
```

Or via Console:

1. Go to Cloud Run → crtlpyme
2. Click “REVISIONS” tab
3. Select previous revision
4. Click “MANAGE TRAFFIC”
5. Set 100% traffic to that revision

Support

For issues or questions:

- Check logs in Cloud Build and Cloud Run
- Review this deployment guide
- Consult GCP documentation: <https://cloud.google.com/run/docs>
- Contact: crtlpyme@gmail.com

Deployment History

| Date | Version | Commit | Status | Notes |
|------------|---------|---------|--------|--------------------------|
| 2025-11-07 | 1.0 | 2dd6ce2 | Ready | Initial deployment setup |

Last Updated: November 7, 2025

Maintained by: CRTLPyme Development Team